

Electromagnetic Spectrum

$$\text{freq of radiation} = f = \frac{c}{\lambda}$$

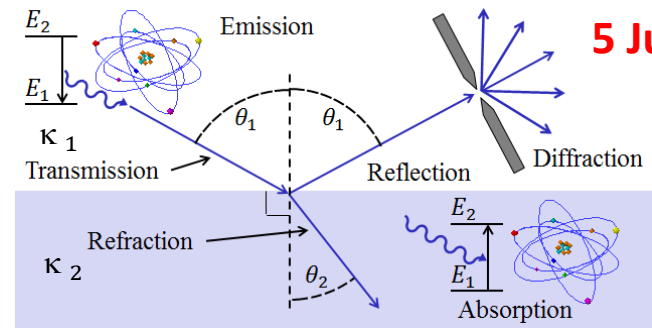
$$\text{period} = T = \frac{1}{f}$$

$$\text{Photon Energy} = E = hf$$

$$v_x = \frac{c}{\kappa_x}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2} = \frac{\kappa_2}{\kappa_1}$$

Gamma	X-Ray	UV	Visible	Infrared	Radio
	30 EHz 10 pm	30PHz 10 nm	750 THz 400 nm	400 THz 750 nm	300 GHz 1 mm

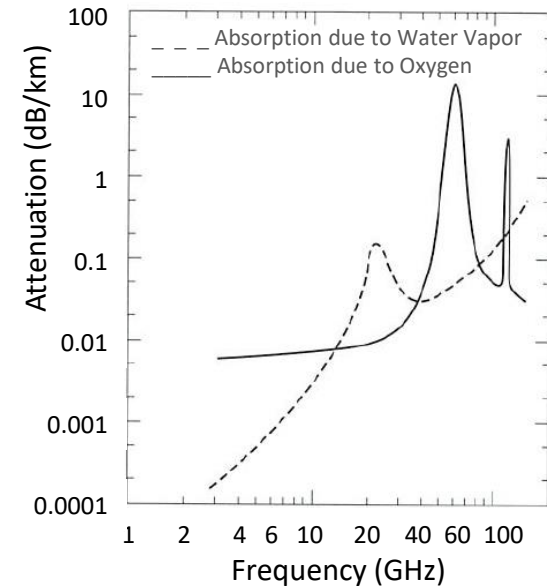
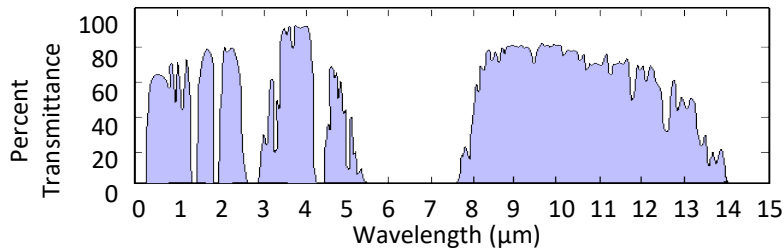


Radio

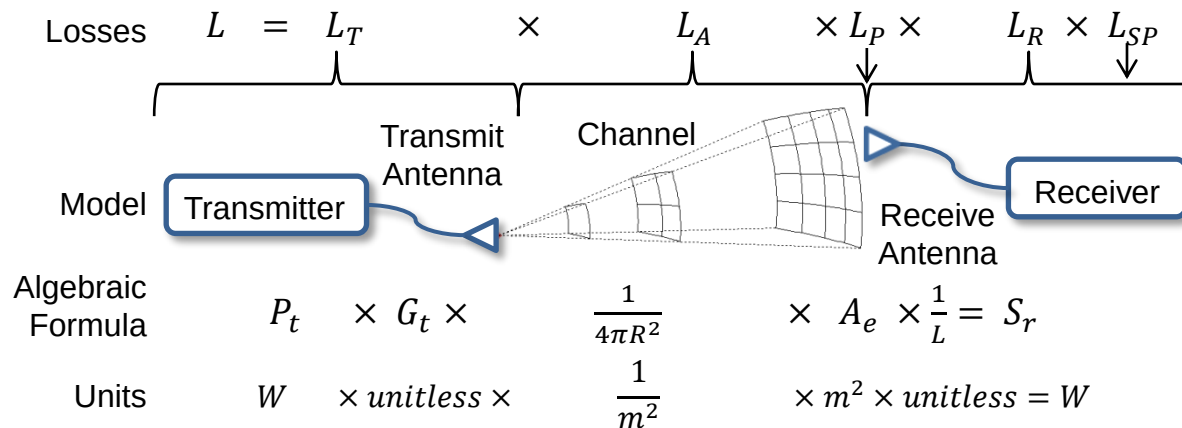
ITU	EHF										SHF					UHF					VHF	HF	MF	LF	VLF	ULF	SLF	ELF	TLF
IEEE	mm	W		V	Ka			K	Ku	X	C		S	L	UHF		VHF	HF	HF										
NATO		M		L	K				J	I	H	G	F	E	D	C	B			A									
Freq	300 GHz	110 GHz	100 GHz	75 GHz	60 GHz	40 GHz	30 GHz	27 GHz	20 GHz	18 GHz	12 GHz	10 GHz	8 GHz	6 GHz	4 GHz	3 GHz	2 GHz	1 GHz	500 MHz	300 MHz	250 MHz	30 MHz	3 MHz	300 kHz	30 kHz	3 kHz	300 Hz	30 Hz	3 Hz
	1.0 mm	2.7 mm	3.0 mm	4.0 mm	5.0 mm	7.5 mm	10 mm	11 mm	15 mm	17 mm	25 mm	30 mm	37 mm	51 mm	75 mm	100 mm	150 mm	300 mm	600 mm	1.0 m	1.2 m	10 m	100 m	1 km	10 km	100 km	10 ³ km	10 ⁴ km	10 ⁵ km
	λ	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	m	m	m	m	km	km	km	km	km	km

Infrared

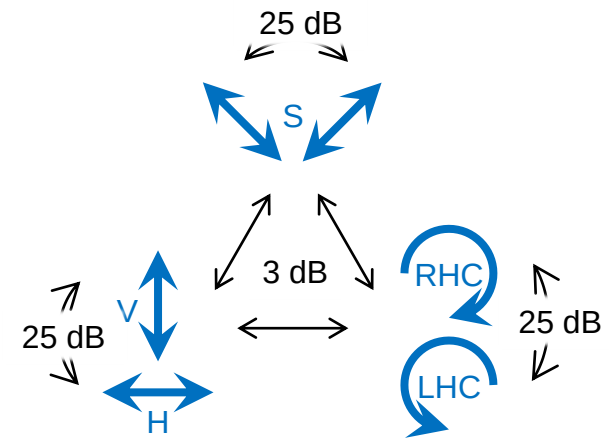
NIR		SWIR		MWIR	LWIR		FIR	
400 THz	300 THz	214 THz	100 THz	37.5 THz	30 THz	20 THz	3 THz	300 GHz
750 nm	1.0 μm	1.4 μm	3 μm	8 μm	10 μm	15 μm	100 μm	1 mm



One-Way Link Equations



Nominal Polarization Losses



Algebraic

$$S_R = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 R^2 L} \quad L = \prod_{i=T}^R L_i \quad A_e = \frac{G_r \lambda^2}{4\pi} = A \eta$$

Logarithmic

$$10 \log(S_r) = 10 \log(P_t) + 10 \log(G_t) + 10 \log(G_r) + 20 \log(\lambda) - 20 \log(4\pi) - 20 \log(R) - 10 \log(L_T) - 10 \log(L_A) - 10 \log(L_P) - 10 \log(L_R) - 10 \log(L_{SP})$$

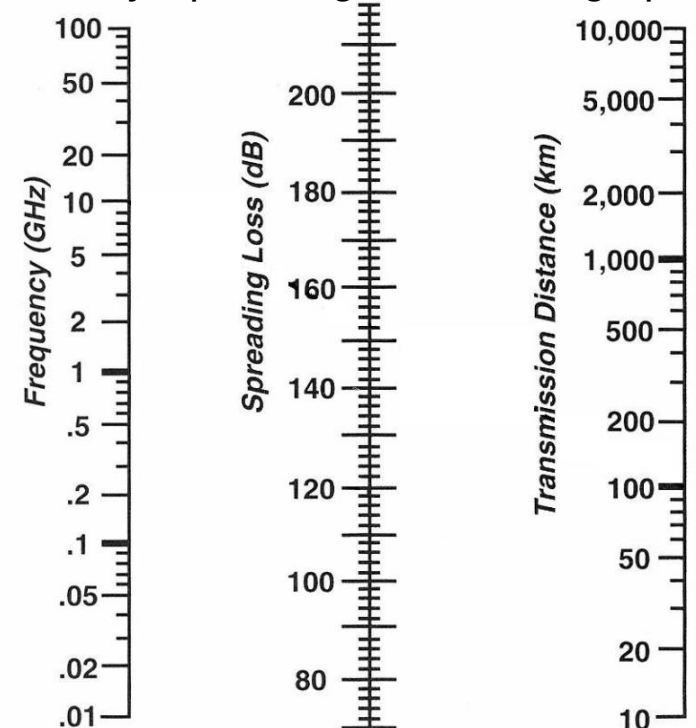
Adamy/dB – units in dB unless otherwise noted

$$S_R(dBW) = P_T(dBW) + G_T(dBi) + G_R(dBi) - L_S(dB) - L_{other}(dB)$$

$$L_S(dB) = 32.4 + 20 \log\left(\frac{Range}{km}\right) + 20 \log\left(\frac{Freq}{MHz}\right)$$

or get L_S from the nomograph

Adamy Spreading Loss Nomograph



Two-Way Link Equations

Monostatic

Algebraic

$$S_R = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 L} \quad L = \prod_{i=T}^R L_i \quad A_e = \frac{G_r \lambda^2}{4\pi} = A\eta$$

Logarithmic

$$\begin{aligned} 10 \log(S_R) \\ = 10 \log(P_T) + 20 \log(G) + 20 \log(\lambda) + 10 \log(\sigma) - 30 \log(4\pi) \\ - 40 \log(R) - 10 \log(L) \end{aligned}$$

$$L = \prod_{i=T}^R L_i$$

Adamy/dB – units in dB unless otherwise noted

$$S_R = P_T + 2G - 103 - 40 \log\left(\frac{Range}{km}\right) - 20 \log\left(\frac{Freq}{MHz}\right) + 10 \log\left(\frac{\sigma}{m^2}\right) - L_{other}, L_{other} = \sum_{i=T}^R L_i \text{ excluding } L_S$$

Deconstructed/Bistatic

Algebraic

$$S_R = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R_1^2 R_2^2 L} \quad \text{where} \quad L = \prod_{i=T}^R L_i \quad \text{and} \quad A_e = \frac{G_r \lambda^2}{4\pi} = A\eta$$

Logarithmic

$$\begin{aligned} 10 \log(S_R) \\ = 10 \log(P_T) + 10 \log(G_T) + 10 \log(G_R) + 20 \log(\lambda) + 10 \log(\sigma) - 30 \log(4\pi) \\ - 20 \log(R_1) - 20 \log(R_2) - 10 \log(L) \end{aligned}$$

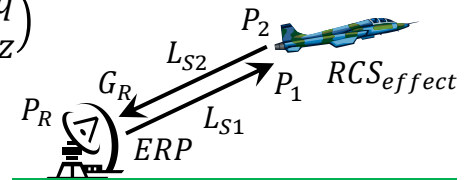
$$L = \prod_{i=T}^R L_i$$

Alternate Adamy Form/dB – units in dB unless otherwise noted

$$ERP = P_T + G_T - L_T \quad L_{S1}(dB) = L_{S2}(dB) = 32.4 + 20 \log\left(\frac{Range}{km}\right) + 20 \log\left(\frac{Freq}{MHz}\right) \text{ or get } L_S \text{ from the nomograph}$$

$$RCS_{effect} = P_2 - P_1 = -39 + 10 \log\left(\frac{RCS}{m^2}\right) + 20 \log\left(\frac{freq}{MHz}\right)$$

$$P_R = ERP - L_{S1} + RCS_{effect} - L_{S2} + G_R - L_{Any other losses}$$



Radar Performance

Antenna

...planar uniform illuminated

...taper illuminated

...SAR

...ESA

Beamwidth rectangular... $\Theta_{3dB} = \frac{.88\lambda}{L}$

Beamwidth circular... $\Theta_{3dB} = \frac{1.02\lambda}{d}$

$\Theta_{3dB} = \frac{1.25\lambda}{d}$

Angular resolution for a... $\rho_A = R\Theta_{3dB}$

Gain for a... $G = \frac{A_e 4\pi}{\lambda^2}$

Phase shift for a ...

—

$L_{eff\ max} = R\Theta_{3dB}$
 $\Theta_{3dB-SAR} = 0.44 \frac{\lambda}{L_{eff}}$

$\rho_{CR(SAR)} = \frac{0.44\lambda R}{L_{eff}}$

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$\Theta_{3dBarray} = \frac{\Theta_{3dB(element)}}{N \cos(\psi)}$

$\rho_A = R\Theta_{3dBarray}$

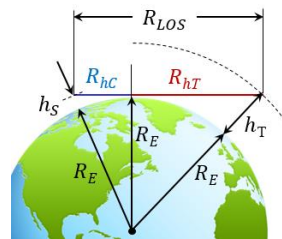
$\Delta\phi = \frac{2\pi d \sin \psi}{\lambda}$ (radians)

Line of Sight

$R_{LOS} = R_{hC} + R_{hT}$ where

$R_{hC} = \sqrt{2k_r R_E h_S + h_S^2}$

and $R_{hT} = \sqrt{2k_r R_E h_T + h_T^2}$



Antenna Far Field

$R_{far} \geq \frac{2d^2}{\lambda}$

Doppler

One-Way

$f_d = -\frac{\dot{R} \cdot f}{c}$

R getting smaller = negative $\dot{R}$ = positive closure velocity = positive Δf

R getting bigger = positive $\dot{R}$ = negative closure velocity = negative Δf

Two-Way

$f_d = -\frac{2\dot{R}}{\lambda}$

$f_d = -\frac{2V_R \cos(\theta)}{\lambda}$

$f_{dMLC} = \frac{2V_R \cos(\theta)}{\lambda}$

$f_{dSLC} = \pm \frac{2V_R}{\lambda}$

Pulse

Pulse Width = τ

PRF = f_r

PRI = T

$f_r = \frac{1}{T}$

$R_u = \frac{c}{2 \cdot f_r}$

$B \approx \frac{1}{\tau}$

Duty Factor = $\frac{\tau}{T} = \tau \cdot f_r$

$B_{LFM} = f_{hi} - f_{low}$

$P_{avg} = P\tau f_r = P \frac{\tau}{T}$

$D_u = \frac{f_r}{2}$

$p_{cr} = \frac{B_{LFM}}{B}$

$t_{int} = NT$

$\rho_R = \frac{c\tau}{2}$

$P_{LFM} = P \cdot p_{cr}$

Receiver/Overall Radar

Receiver noise = kT_0BF

$R = \frac{c\Delta t}{2}$

$P_D = \frac{\text{hits}}{\text{scans}}$

$T_S = T_0F$

$SNR = \frac{NP\tau G^2 \lambda^2 \sigma}{(4\pi)^3 R^4 LkT_0F}$

$SNR = \frac{P_{avg} G^2 \lambda^2 \sigma t_{int}}{(4\pi)^3 R^4 LkT_0F}$

Receiver

system sensitivity = $kT_0BF \cdot SNR$

$\frac{R_{50spec}}{R_{50test}} = \left(\frac{\sigma_{spec}}{\sigma_{test}} \right)^{1/4}$

Electro-Optics

$$E = hf = \frac{hc}{\lambda}$$

(Lens Maker)

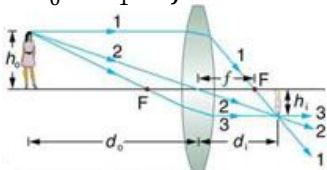
$$\frac{1}{d_0} + \frac{1}{d_1} = \frac{1}{f}$$

$$\sin(\theta) = \frac{1.22\lambda}{D}$$

(Hyperfocal)

$$H = \frac{f^2}{Nc} + f$$

$$N = \frac{f}{D}$$



$$I = I_0 \cos \theta \text{ (Lamberts Cosine)}$$

$$S_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\left(\frac{hc}{\lambda kT}\right)} - 1} \text{ (Planck's Law)}$$

$$\lambda_{peak} = \frac{0.0029}{T} \text{ (Wien Displacement)}$$

$$W = \sigma \epsilon T^4 \text{ (Stefan - Boltzman)}$$

$$\frac{V}{H_{fwd}} = \frac{V}{H_{vert}} \cdot \cos \theta \text{ (Motion Compensation)}$$

$$1 \text{ mr} \approx 1' \text{ at } 1000'$$

$$1 \text{ deg} \approx 1 \text{ NM at } 60 \text{ NM}$$

$$FOV = 2 \tan^{-1} \left(\frac{W}{2f} \right) = 2 \tan^{-1} \left(\frac{\text{Width}/2}{\text{Height}} \right)$$

$$IFOV = 2 \tan^{-1} \left(\frac{X_d}{2f} \right) \quad Q_R = \frac{2BW}{SR} = \frac{GRD}{SR}$$

$$F_s = \frac{1}{2 \cdot IFOV} \cdot (0.001 \text{ cycles/mr})$$

$$F_s = \frac{SR}{2 \cdot BW} \cdot (0.001 \text{ cycles/mr})$$

Data Analysis

Binary Decimal and Hexadecimal

Binary Base 2	Decimal Base 10	Hexadecimal Base 16
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

1553

BIT
TIMES

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----

COMMAND
WORD

		5	1	5	5	1
SYNC	TERMINAL ADDRESS	T/R	SUBADDRESS/MODE	WORD COUNT/ MODE CODE	PAR	

DATA
WORD

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
SYNC		DATA																	PAR

STATUS
WORD

		5					1	1	1	3			1	1	1	1	1	1
SYNC		TERMINAL ADDRESS								RESERVED								PAR
							MESSAGE ERROR			SERVICE REQUEST			INSTRUMENTATION			BROADCAST CMD RECEIVED		
													BUSY			SUBSYSTEM FLAG		
																DYNAMIC BUS ACCEPTANCE		
																TERMINAL FLAG		

7 (DEC) » 00111 (5 bit BIN) 19 (DEC) » 10011 (5 bit BIN) 3 (DEC) » 00011 (5 bit BIN)

SYNC	RTADDRESS					T/R	SUBADDRESS					WORD COUNT					PAR
	0	0	1	1	1	0	1	0	0	1	1	0	0	0	1	1	

HEX Command Word: 3
(4 digit)

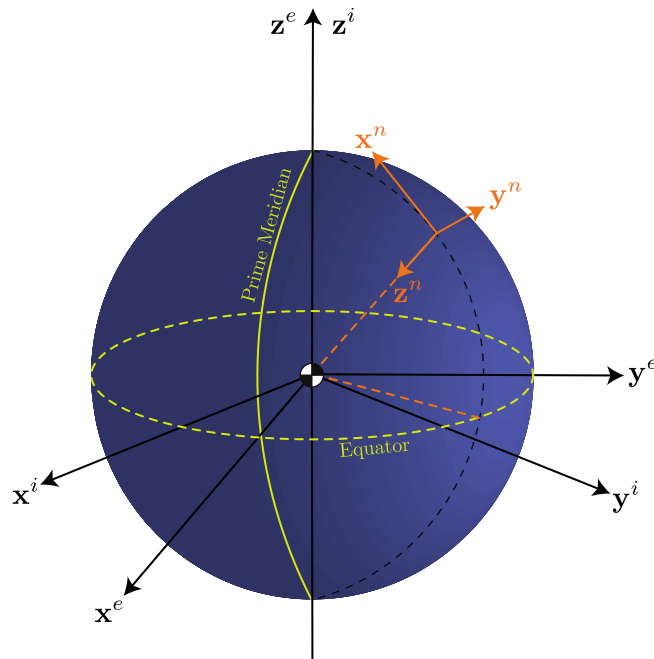
A

6

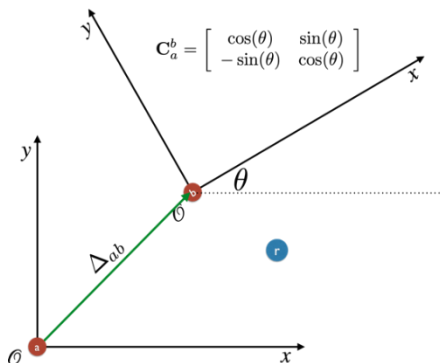
3

Navigation/State Estimation

Reference frames:



Frame Rotations:



$$\begin{aligned} \mathbf{p}_r^b &= \mathbf{C}_a^b (\mathbf{p}_r^a - \Delta_{ab}^a) \\ \mathbf{p}_r^b &= \mathbf{C}_a^b \mathbf{p}_r^a - \mathbf{C}_a^b \Delta_{ab}^a \\ \mathbf{p}_r^b &= \mathbf{C}_a^b \mathbf{p}_r^a - \Delta_{ab}^b \end{aligned}$$

Kalman filter equations:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{F}\mathbf{x}(t) + \mathbf{G}\mathbf{w}(t) \\ \mathbf{w}(t) &\sim \mathcal{N}(0, \mathbf{Q}) \end{aligned}$$

$$\begin{aligned} \mathbf{z}_k &= \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \\ \mathbf{v}_k &\sim \mathcal{N}(0, \mathbf{R}_k) \end{aligned}$$

"Propagate"

$$\begin{aligned} \hat{\mathbf{x}}_{k+1}^- &= \Phi \hat{\mathbf{x}}_k^+ \\ \mathbf{P}_{k+1}^- &= \Phi \mathbf{P}_k^+ \Phi^T + \mathbf{Q}_d \end{aligned}$$

"Update"

$$\begin{aligned} \mathbf{r}_k &= \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^- \\ \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \mathbf{R}_k)^{-1} \\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + \mathbf{K}_k \mathbf{r}_k \\ \mathbf{P}_k^+ &= \mathbf{P}_k^- - \mathbf{K}_k \mathbf{H}\mathbf{P}_k^- \end{aligned}$$

$\mathbf{x}$ is the state

Thing you want to estimate

$\mathbf{z}$ is the measurement

What is directly observable about the system, might not be same as what you estimate

$\mathbf{Q}$ is the process noise

How much error exists on a single step that is unmodeled

$\mathbf{R}$ is the measurement noise

How much error is on the average sensor measurement

$\mathbf{P}$ is the covariance

How much you trust the current estimate

$\mathbf{r}$ is the residuals

How far your estimate is off of the measurement

$\mathbf{K}$ is the Kalman gain

How much you trust measurements

$\mathbf{H}$ is the observation matrix

Maps from state to measurement space

Φ is the propagation matrix

How to move state forward 1 timestep in time

Subscript

X_k indicates state at time k

X_{k+1} indicates state at the next timestep, $k+1$

Superscript

X^- indicates state before a measurement update

X^+ indicates state after a measurement update

Hats

$\hat{X}$ indicates an estimated state

X indicates truth state

But we will never know truth state in reality

Weighted Least Squares:

$$\begin{aligned} \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \underbrace{x_{opt}}_{\beta} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} \\ &\quad \underbrace{\quad}_{\epsilon} \\ \epsilon &\sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_a^2 & 0 \\ 0 & \sigma_b^2 \end{bmatrix}_{\Sigma}\right) \end{aligned}$$

$$\hat{\beta} = (\mathbf{X}^T \mathbf{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{\Sigma}^{-1} \mathbf{y}$$

$$\mathbf{P}_{opt} = (\mathbf{X}^T \mathbf{\Sigma}^{-1} \mathbf{X})^{-1}$$

Constants, Conversions, dB, and Greek

Constant

$c = 299792458 \text{ m/s}$ = speed of light in a vacuum

$e = 2.71828$ = Euler's number

$k = 1.38 \cdot 10^{-23} \text{ Watt second/Kelvin}$ = Boltzmann constant

$h = 6.624 \cdot 10^{-34} \text{ Joule second}$ = Planck's number

$R_e = 6371 \text{ km}$ = Earth's radius

Speed Conversion

From\To	mph	knots	ft/s	km/h	m/s
mph	1	0.868976	1.466666	1.609344	0.44704
knots	1.150779	1	1.687809857	1.852	0.51444444
ft/s	0.681818	0.592484	1	1.09728	0.3048
km/h	0.621371	0.539957	0.911344415	1	0.277777778
m/s	2.236936	1.943844	3.280839895	3.6	1

Distance Conversion

From\To	m	in	ft	st. mile	nau. mile
m	1	39.37008	3.280839895	0.000621371	0.000539957
in	0.0254	1	0.083333333	1.57828E-05	1.37149E-05
ft	0.3048	12	1	0.000189394	0.000164579
st. mile	1609.344	63360	5280	1	0.868976242
nau. mile	1852	72913.39	6076.115486	1.150779448	1

Decibels

Decibels

Ratio dB

0.001 -30

0.01 -20

0.10 -10

0.20 -7

0.50 -3

1.00 0

1.26 1

1.58 2

2.00 3

2.51 4

3.16 5

3.98 6

5.01 7

6.31 8

7.94 9

10.00 10

100.00 20

1000.00 30

$$dB = 10 \log_{10}(\text{ratio})$$

$$\text{ratio} = 10^{\left(\frac{dB}{10}\right)}$$

Power Ratios

$$dB = 10 \log\left(\frac{P_2}{P_1}\right) = 20 \log\left(\frac{V_2}{V_1}\right)$$

$$dBW = 10 \log\left(\frac{\text{Power in } W}{1 W}\right)$$

$$dBm = 10 \log\left(\frac{\text{Power in } mW}{1 mW}\right)$$

$$dBm = 30 + dBW$$

Antenna Gain

$$dBi = 10 \log\left(\frac{\text{Antenna Gain}}{1}\right)$$

RCS

$$dBsm = 10 \log\left(\frac{\text{Target Effective Area (m}^2\text{)}}{1 \text{ m}^2}\right) = 10 \log(\sigma)$$

Angle Conversion

$$\text{deg} = \text{rad} \cdot \frac{180}{\pi}$$

Metric Prefixes

prefix	10 ^x
pico (p)	-12
nano (n)	-9
micro (μ)	-6
milli (m)	-3
-	0
kilo (k)	3
mega (M)	6
giga (G)	9
tera (T)	12
peta (P)	15
exa (E)	18
zetta (Z)	21
yotta (Y)	24

Greek

A	α	alpha
B	β	beta
Γ	γ	Gamma
Δ	δ	delta
E	ε	epsilon
Z	ζ	zeta
H	η	eta
Θ	θ	theta
I	ι	iota
K	κ	kappa
Λ	λ	lambda
M	μ	mu
N	ν	nu
Ξ	ξ	xi (ksi)
O	ο	omicron
Π	π	pi
P	ρ	rho
Σ	σς	sigma
T	τ	tau
Υ	υ	upsilon
Φ	φ	phi
X	χ	chi (kai)
Ψ	ψ	psi
Ω	ω	omega

Joint Electronics Type Designation System

1st Letter - Installation

A – Piloted Aircraft

B – Underwater Mobile (submarine)

C – Cryptographic Equipment (NSA use only)

D – Pilotless Carrier (drone, UAV)

F – Fixed Ground

G – General Ground Use

K - Amphibious

M – Ground Mobile

P – Human Portable

S – Water (surface ship)

T – Transportable (ground)

U – General Utility (multi use)

V – Vehicle (ground)

W – Water Surface & Underwater combined

Z – Piloted/Pilotless Airborne Vehicles combined

2nd Letter – Type of Equipment

A – Invisible Light, Heat Radiation

B – Comsec (NSA use only, was pigeon)

C – Carrier

D – Radiac

E – Laser

F – Fiber Optics

G – Telegraph or Teletype

I – Interphone and Public Address

J – Electromechanical or inertial wire covered

K – Telemetry

L – Countermeasures

M – Meteorological

N – Sound in Air

P – Radar

Q – Sonar and Underwater Sound

R – Radio

S – Special or Combination

T – Telephone (wire)

V – Visual, visible light

W – Armament (not otherwise covered)

X – Fax or Television

Y – Data Processing

Z – Communications (NSA use only)

3rd Letter - Purpose

A – Auxiliary Assembly

B – Bombing

C – Communication (two way)

D – Direction Finding

E – Ejection and/or Release

G – Fire Control or Searchlight Directing

H – Recording and/or Reproducing

K - Computing

L – was Searchlight Control, now “G”

M – Maintenance of Test

N – Navigation Aid

P – was Reproducing, now “H”

Q – Special or Combination

R – Receiving or Passive Detecting

S – Detecting Range and Bearing, Search

T - Transmitting

W – Automatic Flight or Remote Control

X – Identification or Recognition

Y – Surveillance and Control

Z – Secure (NSA use only)

Bold items are most frequently used

Weapons

Bombs

	Type or Guidance							Family
Weight	Training	Dumb	Laser	Laser/GPS	GPS	Tri-mode	EO/IR	
10	BDU-38/BDU-48	-	-	-	-	-	-	
25	BDU-33	-	-	-	-	-	-	
250	-	Mk-81	-	-	GBU-39	GBU-53	-	
500	BDU-50	Mk-82	GBU-12	GBU-54	GBU-38	-	-	
1000	-	Mk-83	GBU-16	-	GBU-32/35	-	-	
2000	BDU-56	Mk-84	GBU-10	GBU-24	EGBU-24	GBU-31	-	GBU-15
5000	-	-	-	GBU-28	EGBU-28	-	-	-

Paveway II
Paveway III
JDAM
SDB

Air Intercept Missiles

Name	Guidance
AIM-9L/M Sidewinder	IR
AIM-9X Sidewinder	IR FPA
AIM-120 AMRAAM	Active Radar

Cluster Bomb Unit

Name	Sub-Munitions
CBU-87 CEM	Bombs
CBU-89 GATOR	Mines
CBU-97 SFW	Smart Bombs
CBU-103 WCMD	Bombs
CBU-104 WCMD	Mines
CBU-105 WCMD	Smart Bombs

Air to Ground Missiles

Name	Guidance
AGM-65 Maverick	(A,B,H,K) TV, (D,F,G) IR, (E) Laser
AGM-84K SLAMMER	GPS/INS Terminal IR
AGM-86C/D CALCM	GPS/INS
AGM-88B/C HARM	Homing
AGM-114 Hellfire	(K,M,N) Laser (L) mmw Radar
AGM-154 JSOW	GPS/INS
AGM-158 JASSM	GPS/INS